

Terraform Task-2

Task Description:

Create 2 EC2 instances on 2 different regions and install nginx using terraform script.

AWS CLI configured :

```
ubuntu@ip-172-31-39-36:~$ aws configure
AWS Access Key ID [None]: 
AWS Secret Access Key [None]: 
Default region name [None]: us-east-1
Default output format [None]:
```

main.tf

```
[ubuntu@ip-172-31-39-36:~/terraform-multi-region-nginx$ cat main.tf
```

```
terraform {
  required_version = ">= 1.5.0"

  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

# Provider for Region 1
provider "aws" {
  region = var.region_1
}

# Provider for Region 2
provider "aws" {
  alias = "region2"
  region = var.region_2
}

# Latest Amazon Linux 2023 AMI in Region 1
data "aws_ami" "al2023_r1" {
  most_recent = true
  owners      = ["amazon"]

  filter {
    name = "name"
    values = ["al2023-ami-*-x86_64"]
  }
}

# Latest Amazon Linux 2023 AMI in Region 2
data "aws_ami" "al2023_r2" {
  provider = aws.region2
  most_recent = true
  owners = ["amazon"]

  filter {
    name = "name"
    values = ["al2023-ami-*-x86_64"]
  }
}

# Security Group in Region 1
resource "aws_security_group" "nginx_sg_r1" {
  name = "nginx-sg-r1"
  description = "Allow HTTP and SSH"

  ingress {
    description = "HTTP"
    from_port = 80
    to_port = 80
    protocol = "tcp"
    cidr_blocks = [var.allowed_cidr]
  }

  ingress {
    description = "SSH"
  }
}
```

variable.tf

```
[ubuntu@ip-172-31-39-36:~/terraform-multi-region-nginx$ cat variables.tf
}
variable "region_1" {
  description = "Primary AWS region"
  type        = string
}

variable "region_2" {
  description = "Secondary AWS region"
  type        = string
}

variable "instance_type" {
  description = "EC2 instance type"
  type        = string
  default     = "t2.micro"
}

variable "allowed_cidr" {
  description = "CIDR allowed for HTTP/SSH (use your IP/32 for better security)"
  type        = string
  default     = "0.0.0.0/0"
}
```

terraform.tfvars :

```
ubuntu@ip-172-31-39-36:~/terraform-multi-region-nginx$ cat terraform.tfvars
region_1      = "us-east-1"
region_2      = "us-west-2"
instance_type = "t2.micro"
allowed_cidr  = "0.0.0.0/0"
```

output.tf :

```
allowed_cidr  = "0.0.0.0/0"
ubuntu@ip-172-31-39-36:~/terraform-multi-region-nginx$ cat outputs.tf
output "region_1_public_ip" {
  value = aws_instance.nginx_r1.public_ip
}

output "region_2_public_ip" {
  value = aws_instance.nginx_r2.public_ip
}

output "region_1_url" {
  value = "http://${aws_instance.nginx_r1.public_ip}"
}

output "region_2_url" {
  value = "http://${aws_instance.nginx_r2.public_ip}"
}
```

```
ubuntu@ip-172-31-39-36:~/terraform-multi-region-nginx$ terraform init
Initializing the backend...
```

Initializing provider plugins...

- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)

Terraform has created a lock file **.terraform.lock.hcl** to record the provider selections it made above. Include this file in your version control repository so that Terraform can guarantee to make the same selections by default when you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
[ubuntu@ip-172-31-39-36:~/terraform-multi-region-nginx$
```

```
[ubuntu@ip-172-31-39-36:~/terraform-multi-region-nginx$
```

```
ubuntu@ip-172-31-39-36:~/terraform-multi-region-nginx$ terraform fmt
```

```
ubuntu@ip-172-31-39-36:~/terraform-multi-region-nginx$ terraform validate
```

Success! The configuration is valid.

```
ubuntu@ip-172-31-39-36:~/terraform-multi-region-nginx$ terraform plan
data.aws_ami.al2023_r1: Reading...
data.aws_ami.al2023_r2: Reading...
data.aws_ami.al2023_r1: Read complete after 2s [id=ami-02e8a1f9718e031b6]
data.aws_ami.al2023_r2: Read complete after 3s [id=ami-0864325d397daa984]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# aws_instance.nginx_r1 will be created
+ resource "aws_instance" "nginx_r1" {
  + ami                        = "ami-02e8a1f9718e031b6"
  + arn                      = (known after apply)
  + associate_public_ip_address = true
  + availability_zone         = (known after apply)
  + cpu_core_count            = (known after apply)
  + cpu_threads_per_core      = (known after apply)
  + disable_api_stop          = (known after apply)
  + disable_api_termination   = (known after apply)
  + ebs_optimized              = (known after apply)
  + enable_primary_ipv6       = (known after apply)
  + get_password_data         = false
  + host_id                   = (known after apply)
  + host_resource_group_arn    = (known after apply)
  + iam_instance_profile       = (known after apply)
  + id                        = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle        = (known after apply)
  + instance_state             = (known after apply)
  + instance_type              = "t2.micro"
  + ipv6_address_count         = (known after apply)
  + ipv6_addresses             = (known after apply)
  + key_name                   = (known after apply)
  + monitoring                 = (known after apply)
  + outpost_arn                = (known after apply)
  + password_data              = (known after apply)
  + placement_group            = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns                = (known after apply)
  + private_ip                 = (known after apply)
  + public_dns                 = (known after apply)
  + public_ip                  = (known after apply)
  + secondary_private_ips      = (known after apply)
  + security_groups            = (known after apply)
  + source_dest_check          = true
  + spot_instance_request_id   = (known after apply)
  + subnet_id                  = (known after apply)
  + tags                       = {
    + "Name" = "nginx-ec2-r1"
  }
  + tags_all                   = {
    + "Name" = "nginx-ec2-r1"
  }
  + tenancy                    = (known after apply)
  + user_data                  = "41d8375350df06970b8bf5be272515d62600caa0"
  + user_data_base64           = (known after apply)
  + user_data_replace_on_change = false
  + vpc_security_group_ids     = (known after apply)
```

```
[ubuntu@ip-172-31-39-36:~/terraform-multi-region-nginx$ terraform apply -auto-approve
data.aws_ami.al2023_r1: Reading...
data.aws_ami.al2023_r2: Reading...
data.aws_ami.al2023_r1: Read complete after 2s [id=ami-02e8a1f9718e031b6]
data.aws_ami.al2023_r2: Read complete after 2s [id=ami-0864325d397daa984]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_instance.nginx_r1 will be created
+ resource "aws_instance" "nginx_r1" {
  + ami                        = "ami-02e8a1f9718e031b6"
  + arn                       = (known after apply)
  + associate_public_ip_address = true
  + availability_zone          = (known after apply)
  + cpu_core_count             = (known after apply)
  + cpu_threads_per_core       = (known after apply)
  + disable_api_stop           = (known after apply)
  + disable_api_termination    = (known after apply)
  + ebs_optimized              = (known after apply)
  + enable_primary_ipv6        = (known after apply)
  + get_password_data          = false
  + host_id                    = (known after apply)
  + host_resource_group_arn     = (known after apply)
  + iam_instance_profile       = (known after apply)
  + id                         = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle         = (known after apply)
  + instance_state             = (known after apply)
  + instance_type              = "t2.micro"
  + ipv6_address_count         = (known after apply)
  + ipv6_addresses             = (known after apply)
  + key_name                   = (known after apply)
  + monitoring                 = (known after apply)
  + outpost_arn                = (known after apply)
  + password_data              = (known after apply)
  + placement_group            = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns                = (known after apply)
  + private_ip                 = (known after apply)
  + public_dns                 = (known after apply)
  + public_ip                  = (known after apply)
  + secondary_private_ips       = (known after apply)
  + security_groups            = (known after apply)
  + source_dest_check          = true
  + spot_instance_request_id    = (known after apply)
  + subnet_id                  = (known after apply)
  + tags                       = {
    + "Name" = "nginx-ec2-r1"
  }
  + tags_all                   = {
    + "Name" = "nginx-ec2-r1"
  }
}
```

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

Search [Option+S]

United States (N. Virginia)

Official (555431941191)

Official-console-user

Instances (1/1) Info

Last updated 31 minutes ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

< 1 >

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☒	nginx-ec2-r1	i-09516ba60bb1777d8	Running	t3.micro	Initializing	us-east-1c	ec2-54-167-19-236.co...

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#Instances:

[Option+S]

United States (Oregon)

Official (555431941191)

Official-console-user

Instances (1/1) Info

Last updated 30 minutes ago

Connect

Instance state

Actions

Launch instances

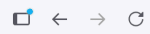
Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

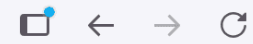
< 1 >

Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elasti
i-0758b91e4eb283eee	Running	t3.micro	Initializing	us-west-2a	ec2-35-161-65-132.us-...	35.161.65.132	-



Not Secure http://54.167.19.236

Nginx from us-east-1



Not Secure http://35.161.65.132

Nginx from us-west-2